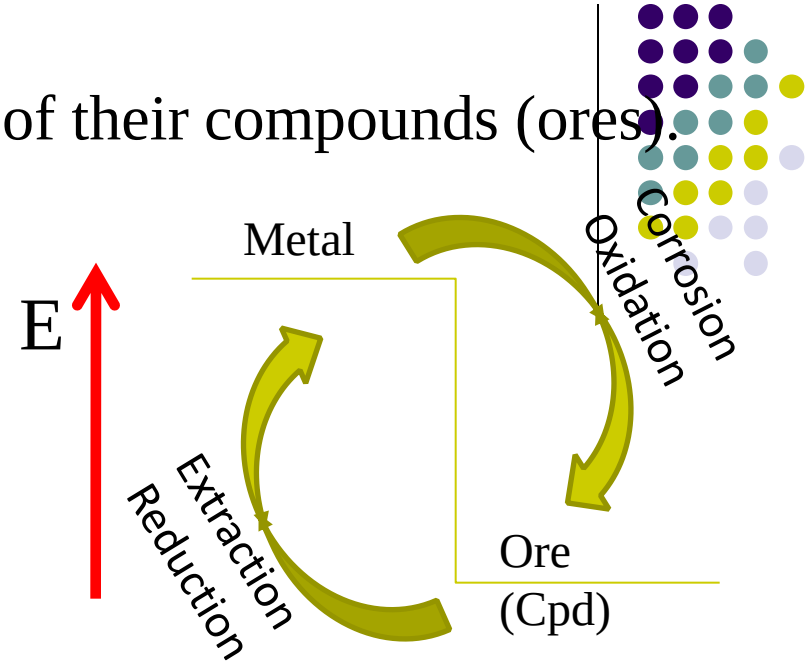
Chapter 3

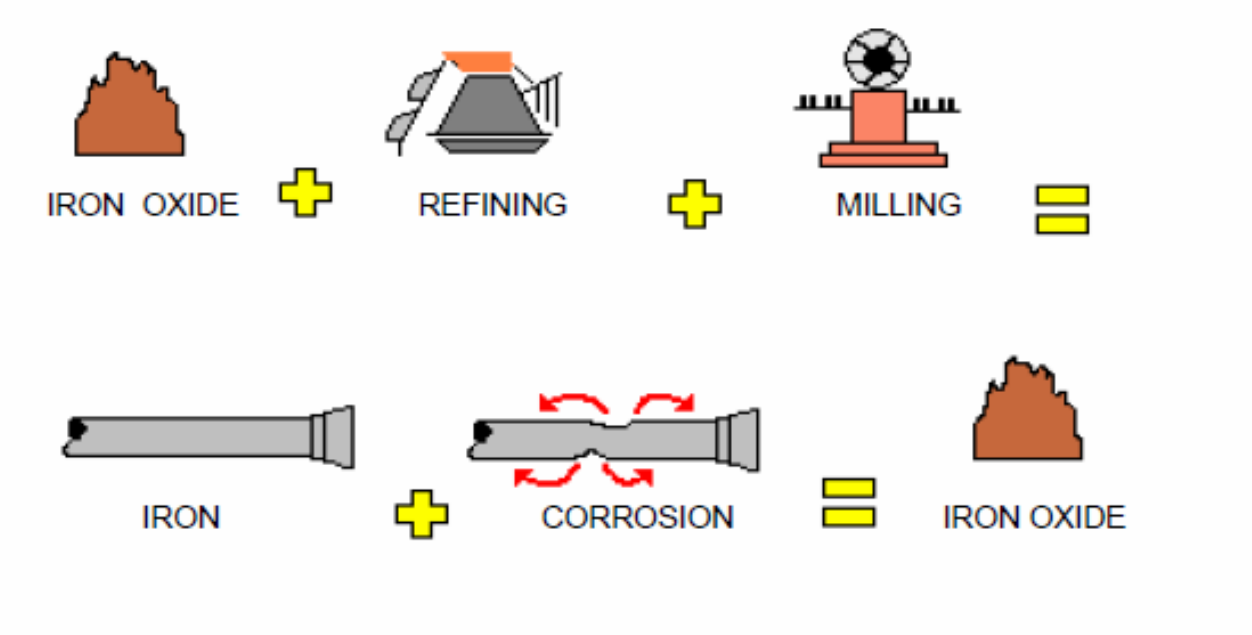
Principles of corrosion

START
STOP

- Metals occur in nature in the form of their compounds (ores).

∴ Metals are less stable than their compounds.





Corrosion

- Corrosion may be defined as destruction of a material due to the contact with its surrounding environment.

- **Types of Corrosion**

1. Direct or Chemical Corrosion **(Dry Corrosion)**.

2. Indirect or Electrochemical Corrosion **(Wet Corrosion)**.

CLASSIFICATION OF CORROSION

Chemical corrosion (dry corrosion)

It is a direct chemical reaction occurs between metal or alloy surface and environment

Takes place by direct combination of metal and its environment . There is no transport of electric charge.

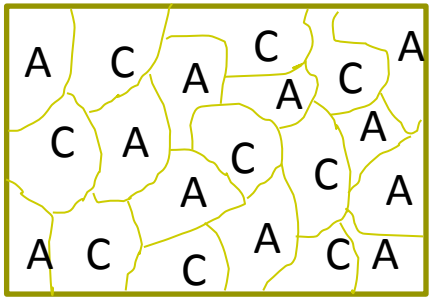
ex:

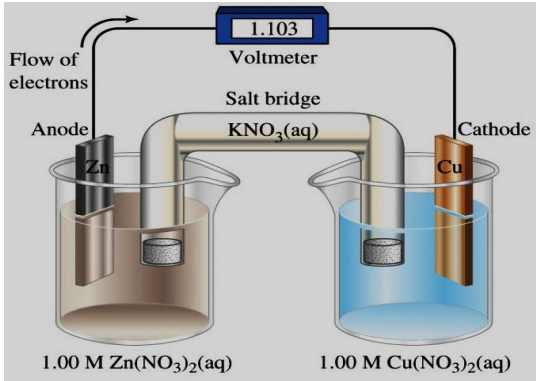
- Dissolution of metals in contact with non-conducting organic media.
- Destruction of metals attacked at high temperatures by aggressive gases.

Electrochemical corrosion (wet corrosion)

It is an electrochemical reaction between metals and the environment.

Takes place at the anodes of the micro electrochemical corrosion cells formed on the metal surface





The five essential components of the corrosion cell are:

1- Anodic zones (A)

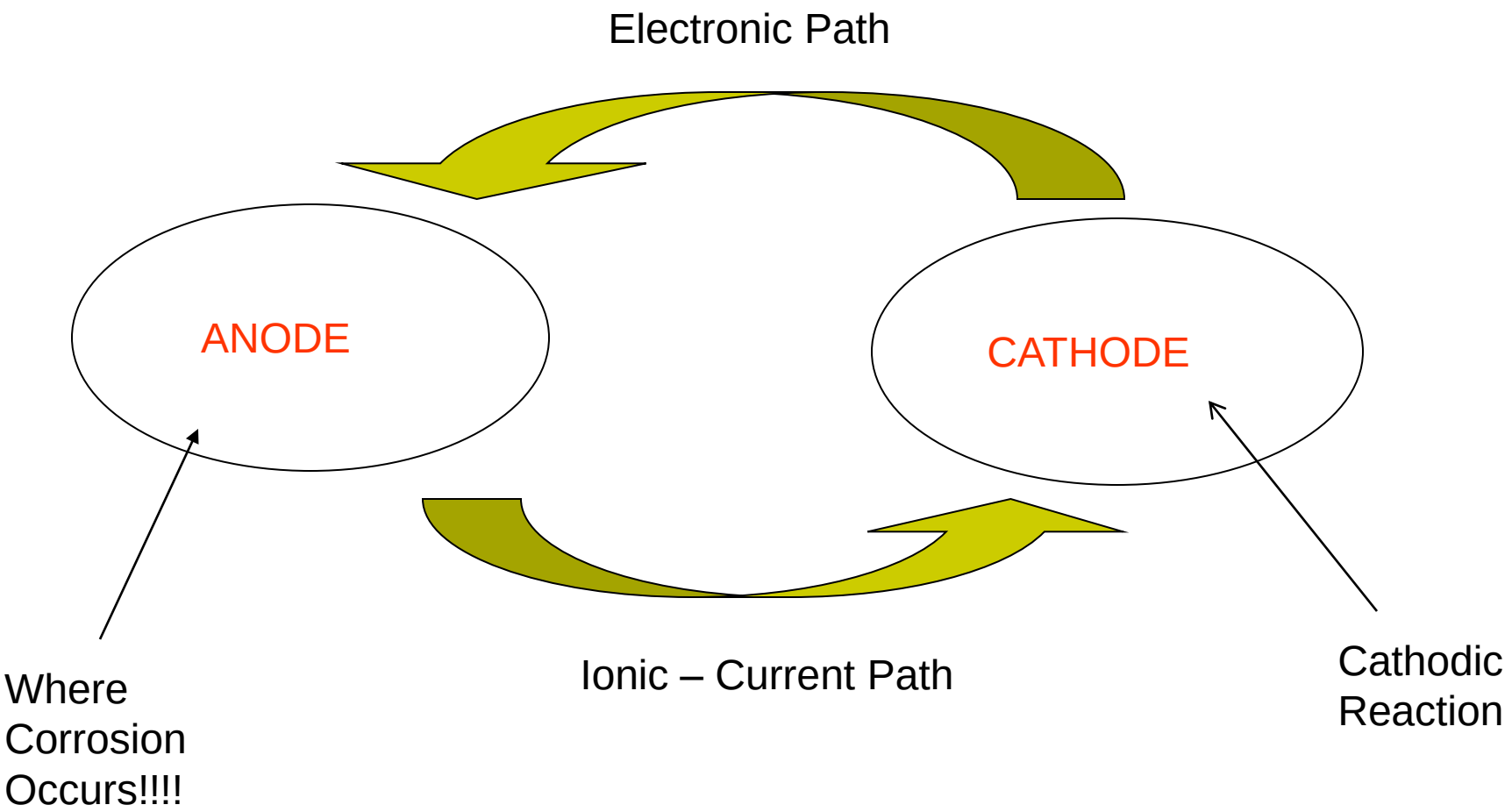
2-Cathodic zones (C)

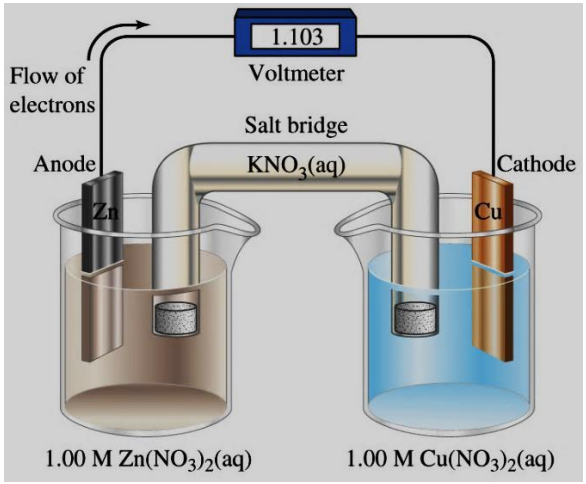
3-electrical contact between them

4-an (ionically) conducting solution

5- cathodic reactant

Requirements for Electrochemical Corrosion:





Oxidation

- LOSS of electrons ($Zn \rightarrow Zn^{2+} + 2e$)
- Removal of H or gain of O
- Take place at anode (dissolution)
- Oxidation Reaction ≅ Anodic Reaction
- Oxidized metal carry a more positive charge (less negative)
- Metal → Positive Ion + Electron(s)
- $\mathbf{M \rightarrow M^{n+} + ne^-}$ (n=1,2,3,…)
- $\mathbf{Na \rightarrow Na^+ + e^-}$ & $\mathbf{Fe \rightarrow Fe^{2+} + 2e^-}$ & $\mathbf{Al \rightarrow Al^{3+} + 3e^-}$

Reduction

- GAIN of electrons ($Cu^{2+} + 2e \rightarrow Cu\downarrow$)
- Removal of O or gain of H
- Take place at cathode (deposition)
- Reduction Reaction ≅ Cathodic Reaction
- Reduced species carried a more negative charge (less positive).
- For strong acidic medium (pH<4)

  $\mathbf{2H^+ + 2e^- \rightarrow H_2}$
- For neutral or basic medium (pH= 4-10)
- $\mathbf{1/2\ O_2 + H_2O + 2e^- \rightarrow 2OH^-}$

The corrosion cell reactions

- ***The anodic reaction is:***

$M \rightarrow M^{+z} + ze$

- ***The most common cathodic reactions are:***

(I) $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$ $e^o = 1.23$ V O_2 rich acidic media

OR

(II) $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$ $e^o = 0.401$ V O_2 rich non acidic media

OR

(III) $2H^+ + 2e^- \rightarrow H_2$ $e^o = 0.0$ V O_2 poor acidic media

OR

(IV) $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$ $e^o = -0.828$ V O_2 poor non acidic media

pH-scale

It’s a scale to determine the acidity or alkalinity strength.

0 → 4	4 → 7	7 → 10	10 → 14
Strong acid	Weak acid	Weak base	Strong base
Sulfuric acid Hydrochloric acid Phosphoric acid	Acetic acid Oxalic acid Citric acid	Ammonia	Sodium hydroxide Potassium hydroxide

Electrochemical Cells

Galvanic (Voltaic) Cell

generating electricity by using a spontaneous chemical reaction (one with ΔG = - ve)

Electrolytic Cell

using electricity to force a chemical reaction to occur (one that is *non-spontaneous*, ΔG = + ve).

The two electrodes are connected with an external source of potential..

Electrochemical cells

Chemical Energy $\xrightarrow{\text{Galvanic cell}}$ **Electrical Energy**

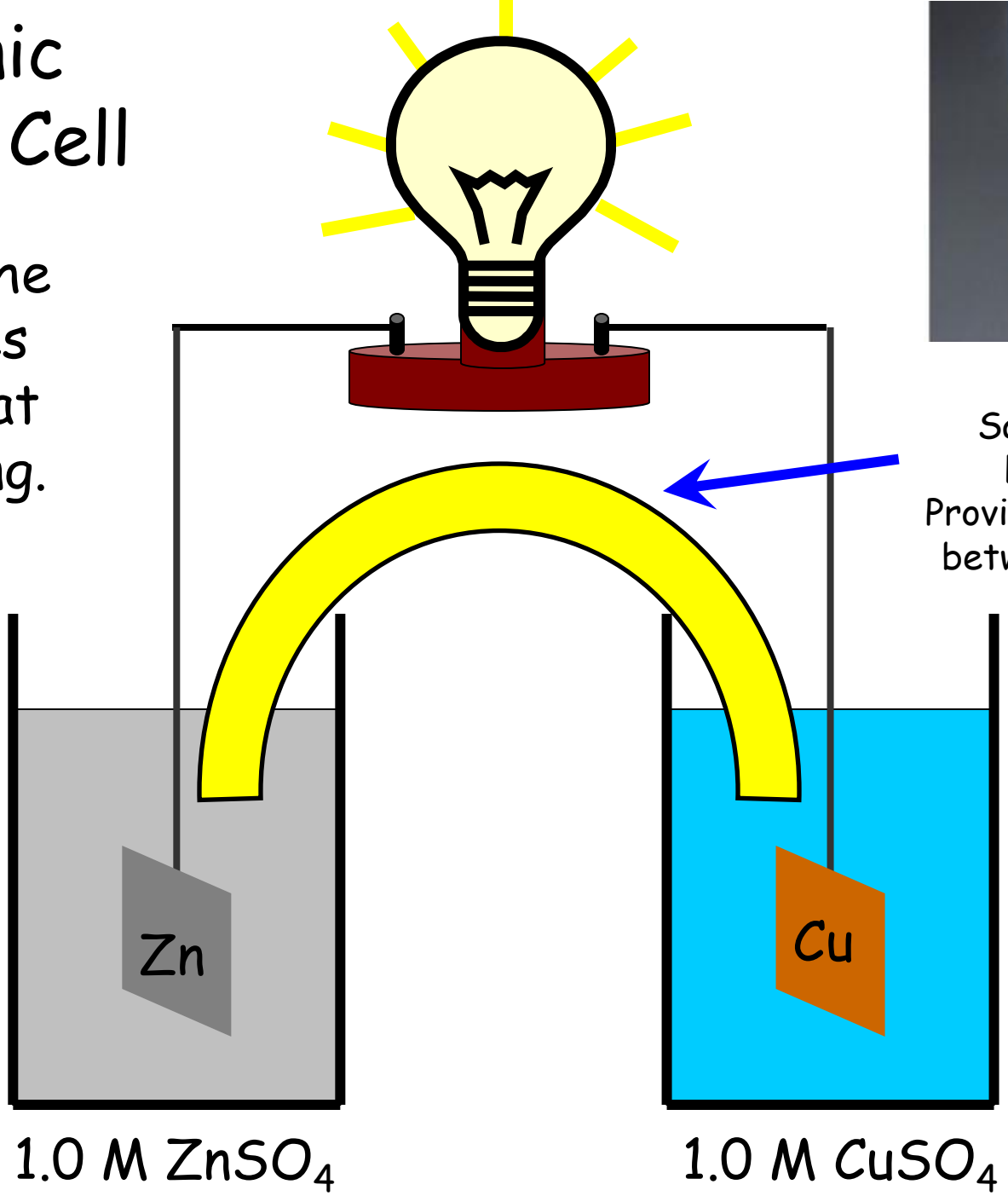
Chemical Energy $\xleftarrow{\text{Electrolytic cell}}$ **Electrical Energy**

Corrosion cell

- **Galvanic corrosion cell**
  - **concentration corrosion cell**
- **Electrolytic corrosion cell**

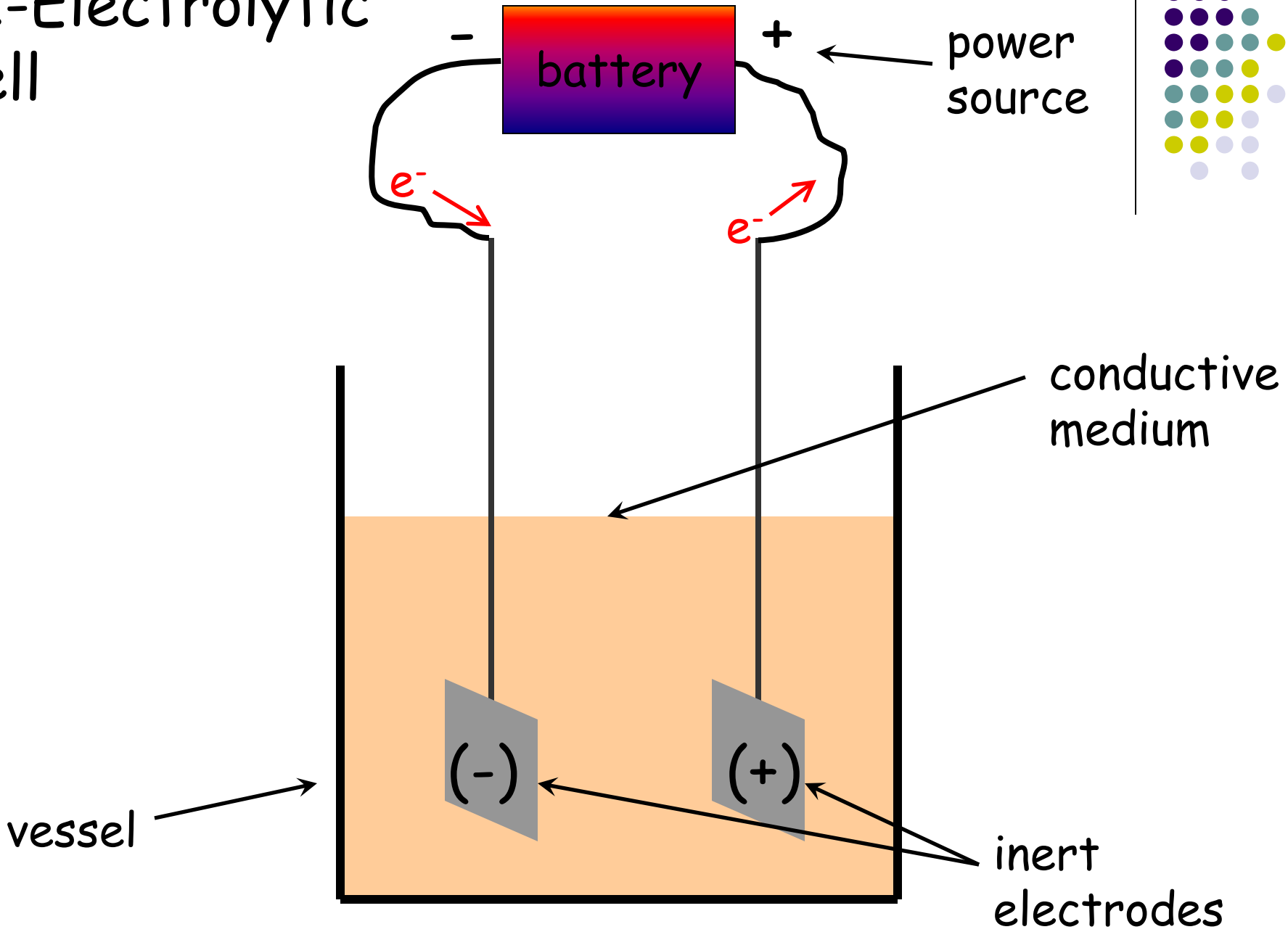
I- Galvanic (Voltaic) Cell

Observe the electrodes to see what is occurring.



Salt bridge – KCl in agar Provides conduction between half-cells

II-Electrolytic Cell



Sign or polarity of electrodes

Remember

The corrosion cell reactions

- ***The anodic reaction is:***

$M \rightarrow M^{+z} + ze$

- ***The most common cathodic reactions are:***

(I) $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$ $e^o = 1.23$ V O_2 rich acidic media

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OR

(IV) $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$ $e^o = -0.828$ V O_2 poor non acidic media

The Electrochemical Series

Table 16.1 The Standard emf Series

	Electrode Reaction	Standard Electrode Potential, V^0(V)
	$Au^{3+} + 3e^- \longrightarrow Au$	+1.420
	$O_2 + 4H^+ + 4e^- \longrightarrow 2H_2O$	+1.229
	$Pt^{2+} + 2e^- \longrightarrow Pt$	~+1.2
	$Ag^+ + e^- \longrightarrow Ag$	+0.800
Increasingly inert (cathodic)	$Fe^{3+} + e^- \longrightarrow Fe^{2+}$	+0.771
	$O_2 + 2H_2O + 4e^- \longrightarrow 4(OH^-)$	+0.401
	$Cu^{2+} + 2e^- \longrightarrow Cu$	+0.340
	$2H^+ + 2e^- \longrightarrow H_2$	0.000
	$Pb^{2+} + 2e^- \longrightarrow Pb$	−0.126
	$Sn^{2+} + 2e^- \longrightarrow Sn$	−0.136
	$Ni^{2+} + 2e^- \longrightarrow Ni$	−0.250
	$Co^{2+} + 2e^- \longrightarrow Co$	−0.277
	$Cd^{2+} + 2e^- \longrightarrow Cd$	−0.403
	$Fe^{2+} + 2e^- \longrightarrow Fe$	−0.440
Increasingly active (anodic)	$Cr^{3+} + 3e^- \longrightarrow Cr$	−0.744
	$Zn^{2+} + 2e^- \longrightarrow Zn$	−0.763
	$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$	-0.828
	$Al^{3+} + 3e^- \longrightarrow Al$	−1.662
	$Mg^{2+} + 2e^- \longrightarrow Mg$	−2.363
	$Na^+ + e^- \longrightarrow Na$	−2.714
	$K^+ + e^- \longrightarrow K$	−2.924

$Au^{3+} + 3e^- \leftrightharpoons Au$ $E^0 = 1.420$ V

Strong tendency to occur as a reduction

(i.e. as written $Au^{3+} + 3e^- \rightarrow Au$)

$Na^+ + e^- \leftrightharpoons Na$ $E^0 = -2.71$ V

Strong tendency to occur as an oxidation

(i.e. $Na \rightarrow Na^+ + e^-$)

$O_2 + 4H^+ + 4e^- \leftrightharpoons 2H_2O$ $e^o = 1.23$ V O_2 rich acidic media

$Cu^+ + 1e \leftrightharpoons Cu$ $e^o = 0.52$V

$O_2 + 2H_2O + 4e^- \leftrightharpoons 4OH^-$ $e^o = 0.401$ V O_2 rich non acidic media

2X $Cu^{2+} + 2e \leftrightharpoons Cu$ $e^o = 0.34$V

$2H^+ + 2e^- \leftrightharpoons H_2$ $e^o = 0.0$ V O_2 poor acidic media

$Fe^{3+} + 3e \leftrightharpoons Fe$ $e^o = -0.04$V

$Fe^{2+} + 2e \leftrightharpoons Fe$ $e^o = -0.44$V

$2H_2O + 2e^- \leftrightharpoons H_2 + 2OH^-$ $e^o = -0.828$ V O_2 poor non acidic media

$O_2 + 2H_2O + 2Cu \rightarrow 2Cu^{2+} + 4OH^-$ in O_2 rich non-acidic media

$4H^+ + 2Cu \rightarrow 2H_2O + 2Cu^{2+}$ acidic media

Polarization

the shift of the electrode potential from its equilibrium value

The rate of electrochemical reaction is limited by various physical and chemical factors.

The electrochemical reaction is retarded by several physical and chemical factors.

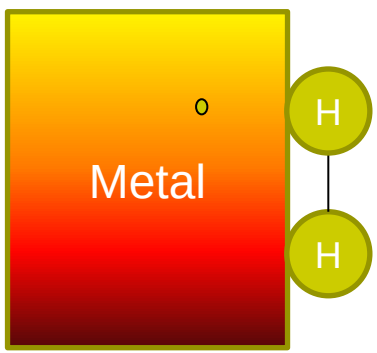
Polarization is inversely proportional to the corrosion rate.

Steps of electrochemical reaction:

1. Ion transfer from the bulk of solution to the surface through the diffusion layer (mass transfer step)
2. Chemical reaction between ions and electrons. (Oxidation or reduction) (charge transfer step)
3. Material transfer from the surface of metal to the bulk of solution through the diffusion layer. (mass transfer step).

Example: Reduction of hydrogen:

$$2H^+ + 2e^- \rightarrow H_{2(g)}$$

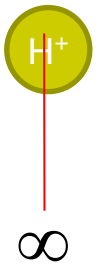


(1) mass transfer step

(2) Charge (electron) transfer step

Chemical reaction

(3) mass transfer step



-the first and third steps are mass transfer steps.
-the second step is chemical reaction step.

Polarization can be conveniently divided into two types:

Concentration polarization

Refers to electrochemical reaction that's controlled by the mass transfer (diffusion) step in the electrolyte.

Condition: the mass transfer step is slower than the chemical reaction.

Occurs in diluted solutions

Activation polarization

Refers to electrochemical reaction that's controlled by the chemical reaction sequence at the metal surface.

Condition: the chemical reaction is slower than the mass transfer.

Occurs in concentrated solutions

Examples:

Corrosion rate in concentrated electrolyte is chemical reaction controlled. (Act. Pol.) slow step 2

Corrosion rate in diluted electrolyte is diffusion controlled. (Conc. Pol.) slow step 1 or 3

Corrosion rate ↑ polarization ↓

Passivity

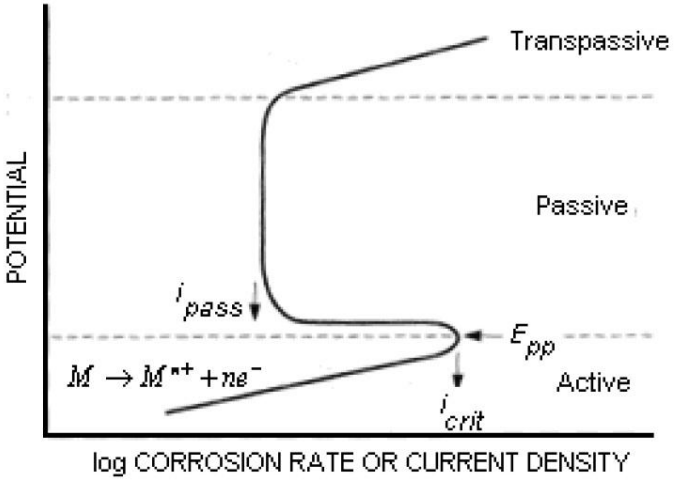
Some metals and alloys under particular environmental condition loss their chemical reactivity and become inert.

Environment	Iron	Aluminum
Air	Non passive	passive
Nitric acid (HNO_3)	passive	Non passive

<u>Passivity</u> *may be defined as the formation of a thin protective film (1-10 nm thickness) on a metal surface due to its reaction with the surrounding environment under oxidizing conditions. (Passive film is protective only if: (1) adherent (2) non-porous, dense and compact)., e.g., the formation of oxide layers of Al_2O_3, TiO_2, Cr_2O_3. this is why Cr is added to Fe to prepare stainless steel*

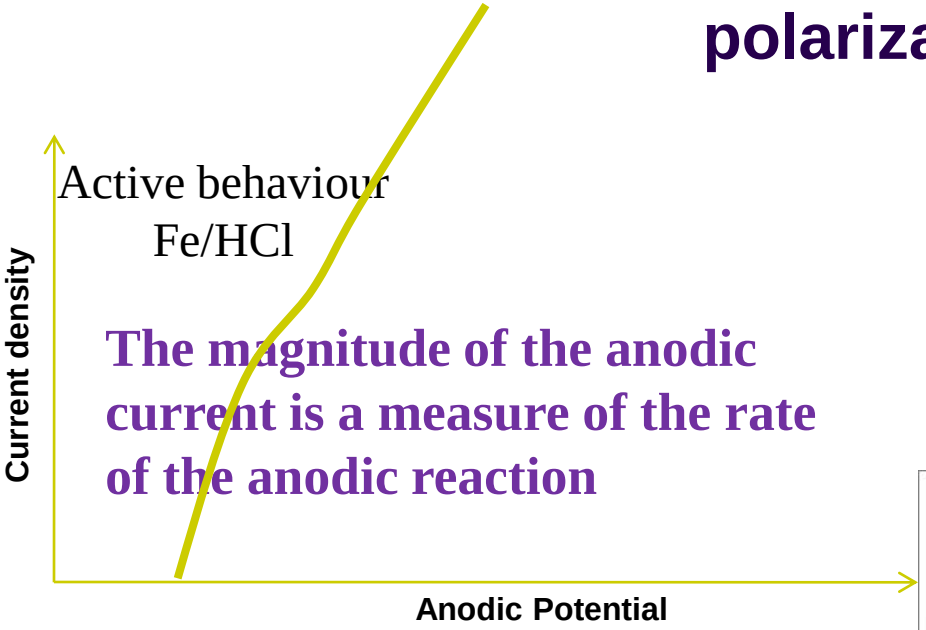
<u>Passivity depends mainly on:</u>

- The nature of the metal or alloy.
- Type of the environment (composition, temp., humidity, ….).

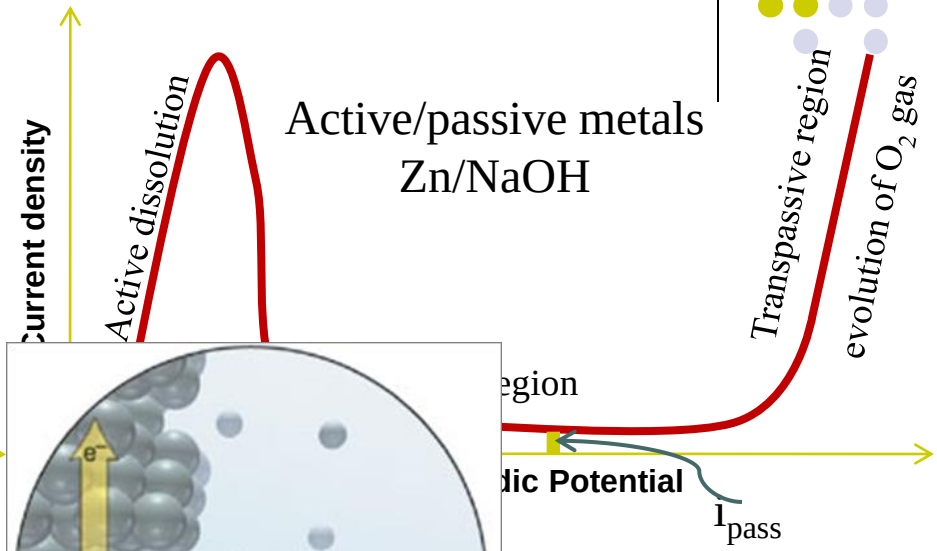


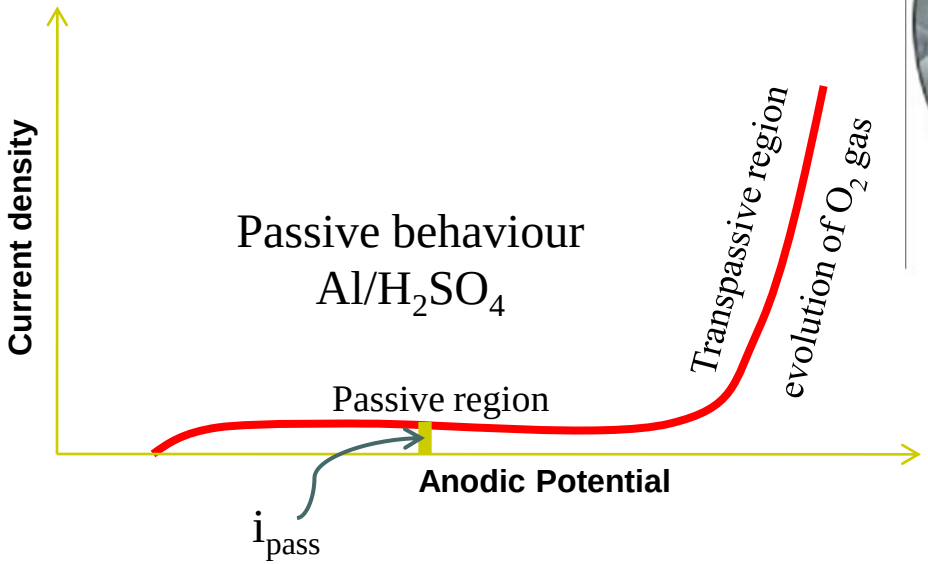
Potentiodynamic polarization Passivity curve

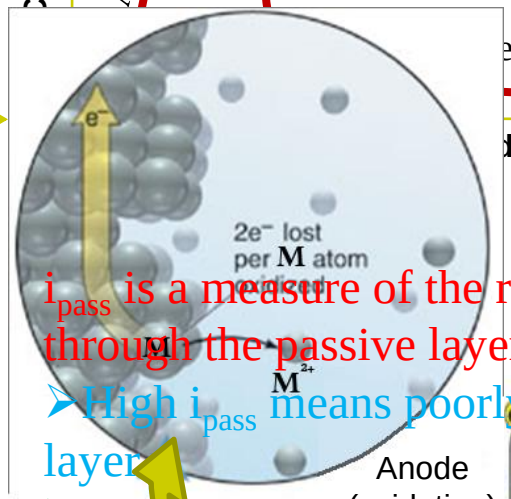
Potentiodynamic polarization



The magnitude of the anodic current is a measure of the rate of the anodic reaction

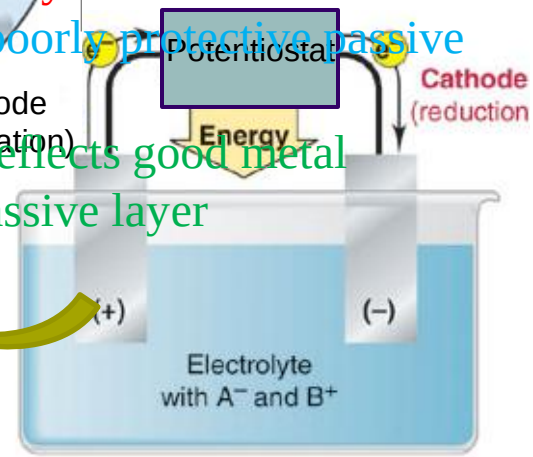


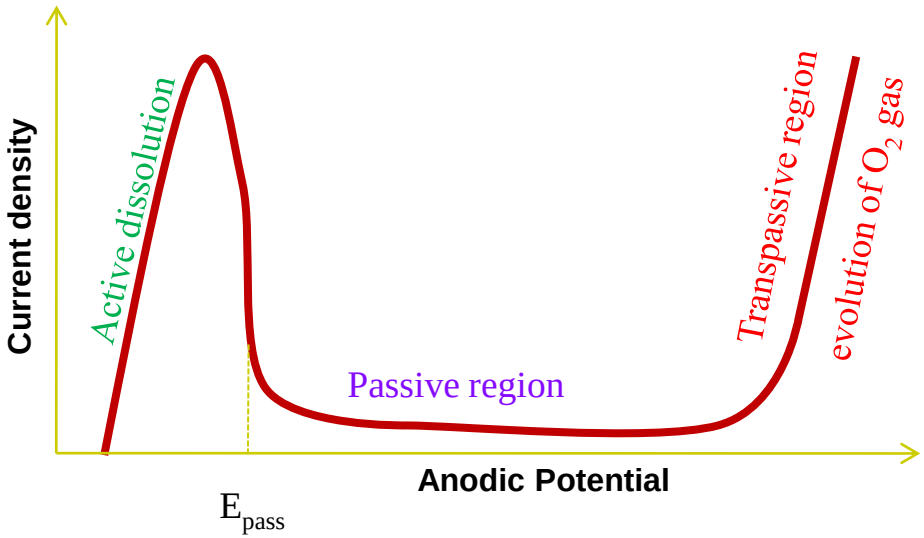




i_{pass} is a measure of the rate of dissolution through the passive layer

- High i_{pass} means poorly protective passive layer
- Small i_{pass} value reflects good metal protection by the passive layer





Active region: in this region, a slight increase in the potential or oxidizing power of the solution causes rapid increase in the corrosion rate.

Passive region: by increasing the oxidizing power the corrosion rate show sudden decrease and further increase in the oxidizing power produce no change(rate is constant).

- Trans passive region: at very high values of oxidizing power, the corrosion rate or the rate of another oxidation process again increases as oxidizing power increase.

Corrosive environment

Air, water, soil, gases, and acids may acts as corrosive environment.

Types of corrosive environment:

- **Air**:
  - Moist and humid air is more corrosive than dry air.
  - Hot air is more corrosive than cold air.
  - Polluted air is more corrosive than fresh air.
- **Water**:
  - Hot water is more corrosive than cold water.
  - Salty water is more corrosive than fresh water.
  - Polluted water is more corrosive than pure water.
- **Gases**:

  Acidic gases (SO_x, NO_x, and H_2S.) are more corrosive than Basic gases (NH_3).
- **Soil**:

  Presence of water, salts, gases, stray current, pollutants, affects the corrosion rate greatly.

- **Acids**:

  Acids are very corrosive environment.

Acid	Mineral	Organic	Fatty
Strength	Very strong	Weak	Very weak
Examples	H_2SO_4, HNO_3, H_3PO_4, HCl……	Acetic and Formic acid	Animals and plant oils

- **Bases:**

  Strong bases (sodium hydroxide, potassium hydroxide) are more corrosive than weak bases (ammonia solution).